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Division of Minerals & Geology

SOUTHDOWN, INC.

ROCKRIDGE QUARRY AT DOWE FLATS

3 YEAR COMPLIANCE REVIEW

TECHNICAL AND ENGINEERING COMPLIANCE

MANAGEMENT AND MONITORING PLANS

AUGUST 30, 1997
Rev: 9/3/97

INTRODUCTION

The County Commissioners approved the Dowe Flats Mine on June 28, 1994 (Resolution No. 94-81 - Special Use No. SU-93-14 and Vacation No. V-93-8). The Board also approved the Development Agreement for the Project on July 12, 1994. These approvals required a compliance review three (3) years from the date of approval. This document addresses current status and compliance with the key provisions in this approval including mining and reclamation, technical and engineering issues, and management and monitoring three (3) years from the time of approval.

TECHNICAL AND ENGINEERING COMPLIANCE

Mining and Reclamation Plans

Mining operations started in late 1996 and early 1997, and are on-going today. Mine pits have been opened in the 1st, 2nd and 3rd ridges. The 4th ridge pit will be opened shortly. Reclamation was not planned to commence until year three of mining, however overburden and topsoil stockpiles are currently being created for use in subsequent reclamation.

The mine pits and stockpiles are located as shown on the permit maps. Production rock has and will continue to be removed from the stockpile areas so that the resource is removed before it is covered. The mining sequence (generally planned to proceed from south to north) is progressing as permitted except for a temporary change in the mining sequence on the 1st ridge. In order to recover limestone from beneath the crusher location, the planned "year two" mine area is being mined first. After the second year of mining, mining will return to the planned sequence.

Interim Stockpiles

A 5.5 acre overburden stockpile has been created south of the 2nd ridge mine pit. This overburden stockpile will continue to grow over the next 3 years to fill this 25 acre triangle. Topsoil is being stockpiled east the mine area and west of the relocated County Road 47.

Operations Center

The Special Use Permit allowed a mine operations center if Southdown received Site Plan Review approval by the Board of County Commissioners. Two such Site Plan Reviews have been approved. The approved components of the operations center are:

- Office trailer -- Complete
- Fueling area -- Under Construction, Est Comp by 9/30/97
- Garage -- Final Design and Bid, Est Comp by 11/30/97
- Shower room -- Final Design and Bid, Est Comp by 11/30/97
- Parking -- Under Construction, Est Comp by 9/30/97

These land uses are located south and southeast of the mining area in the vicinity of the Supply, Highland, and Rough & Ready ditches.

Crusher

A crusher has been constructed at the south end of the 1st ridge within the envelope shown in the permit application. The crusher is set into the limestone ridge, as planned, and is covered by a metal building to control dust. The slopes around the crusher are currently covered with crushed rock. These will be covered with top soil this fall and planted before Spring. This should help this equipment to blend into the terrain.

Conveyor and Highway 66 Crossing

The conveyor belt and tube over State Highway 66 have been built and are now operational. The conveyor and crossing are located in the corridor shown in the permit application.

The visual impact of the crossing structure was carefully reviewed by the County and architectural approvals were required before construction. Several computer simulations of crossing structure styles were reviewed by the County who selected the style and color that is now present. Landscaping plans for the north and south side of Highway 66, at the crossing, were also submitted to and approved by the County (as well as signage). Landscaping is currently being bid and contracted. Current estimates are that nearly \$70,000 worth of trees and woody plants will be placed in this area. Work should be completed this fall.

A Longmont water line along the south side of Highway 66 was a concern because it was close to one of the planned conveyor support foundations. The City did not want the pipeline damaged and required future access to it. This issue was satisfactorily resolved through discussions with the City on the exact location of the foundation, field inspections by the City during construction, and Southdown granting an easement to Longmont for the water line where an easement did not previously exist.

Maximum Acreage of Disturbance

The Special Use Permit limited the area that could be disturbed by mining and reclamation at any one time to a maximum of 120 acres. Once 120 acres of land has been disturbed, revegetation must commence before additional acreage can be disturbed. Currently, less than 60 acres has been disturbed (as measured from very recent air photos and field inspections). This area will increase over the next 3 years to the maximum acreage allowed. At this point the operation will reach steady state and mining activity will advance northward at a pace of about 300 feet per year.

Berming Requirements

Screening berms were required along the relocated County Road 47 to minimize the views of the mining operation. The initial segment of the berm has been constructed at the south end of County Road 47 and it adequately blocks views of the current mining activity. The berm is currently being extended further north. Further extensions will continue over time as mining advances to the north.

Several additional screening berms are being constructed along the southern boundary of the mine near the new County Road and our neighbors along Highway 66. These berms are an attempt at sight and noise abatement. All should be completed this fall and should be planted before the Spring rain.

Current Operations

The major conveyor construction work was completed in early June. From that time until mid-August, only small volumes of rock were crushed and transported due to numerous electrical and mechanical problems with the system. Full operation commenced during the week of August 18, 1997. Since that time, about 6500 tons of rock has been being crushed and transported each scheduled operating day (a total about 45,000 tons for the month). As of the beginning of September, it appears that we are entering steady-state condition with this system. It is anticipated that we will operate the system for 8 to 10 hours per day, 4 or 5 days per week (Monday-Thursday or Friday).

Road Relocations and Vacations

County Road 47

The Special Use Permit allowed relocation of portions of County Road 47 in Dowe Flats. The relocation and a required trail corridor were dedicated to Boulder County in January, 1995. The abandoned portion of County Road 47 has been vacated by the County and now runs through the center of the active quarry.

County Road 49

The Special Use Permit required the demolition of the old concrete bridge across St. Vrain Creek on County Road 49, the vacation of County Road 49 up to the Southdown/Bullock property line, and the construction of a "cul de sac" at the Bullock/Southdown property line. In addition, the vacation of County Road 49 was a precondition for donation of the Indian Mountain property to the County Parks and Open Space. This plan was predicated on the continued viability of the old, green, iron bridge which would provide necessary, legal access to adjacent property owners. After the demolition of this old, iron bridge, Southdown worked with the County Public Works Department to develop a new plan to leave the old concrete bridge in place and, at the same time, not request a vacation of County Road 49. This issue is being addressed in a separate process from this third annual review.

Blasting

Blasting commenced at Dowe Flats in March, 1997. Blast warning and schedule signs have been posted on the new County Road 47. All the requirements of the Special Use Permit pertaining to blasting have been complied with.

Pre-Blast Surveys

Southdown has conducted pre-blast surveys on residences and structures in the area to document their condition before the first blast at Dowe Flats. These reports (including photographs) will protect both the neighboring property owners and Southdown in the event of future damage claims.

Blast Monitoring

Southdown has installed two permanent seismic monitoring stations at Dowe Flats, and also has one portable monitor. The permanent stations are located at the northern and southern ends of the mining area. The monitors measure ground motion due to blasting.

To date, ground motion from all blasts has been well under the permit standards.

Special Monitoring - Montgomery School and Longmont Water Tank

Special standards and monitoring provisions were applied to the Montgomery School and the Longmont Water tank. Very low ground vibration standards (i.e. .25 inches per second) were applied to each structure and these have been complied with (the portable seismic monitor has been used at these locations). The "Montgomery School

Monitoring Report, 1997" by Western Historical Studies, Inc. documents this monitoring. Future blasting will move further from these sites, thus further reducing ground vibrations.

Dust Control

Dust control during construction consisted mainly of watering control using the water wagon to spray roads and construction sites. In addition, the permanent, unpaved roads were treated with Magnesium Chloride to help retain moisture and minimize dust generation due to traffic.

As we enter the production phase of the project, several additional measures are required. Dust is typically generated when the material is crushed or dropped. No dusting typically occurs as crushed rock rides the conveyor. First, there are dust collectors on each transfer point on the crusher-conveyor system. Dust collectors function in a very similar manner to a household vacuum cleaner with a fan and a permeable bag. Several of these dust collectors have already undergone major revisions to make them more effective. Changes such as additional skirting and sealing has occurred to make the vacuum fan more effective in pulling in the fugitive dust particles. These transfer points are now controlling fugitive dust well.

another potential fugitive dust generation point is where the rock is discharged from the stacker belt in to the material stockpile. This unit has been designed to raise and lower across a 50 height to control the drop to between 18 and 24 inches. In addition, a fogging spray has been added at the stacker discharge. This combination appears to be very effective even in dry conditions.

The crusher hopper received a special dust control measure, the large metal building covering its entrance. This building allows a haul truck to back into the hopper to dump its 100 ton payload while containing the dust in the building. It is working very effectively.

Wind Speed Monitoring and High Wind Closure

A recording anemometer is located at the mine site and a read out of wind speed is available in the control room at the cement plant. This anemometer has an alarm tied to it to alert the operator whenever sustained winds exceed 30 mph. When this level is attained, loading and hauling will cease (drilling and conveying can continue). No high wind closures have occurred to date.

Noise

Residential noise standards of 55 decibels are required at the Special Use Permit boundary. Engineering Dynamics Inc. measured noise at the mine

under full operating conditions in July 1997. Noise levels from the operation are well below the required standards.

The Special Use Permit also required Southdown to seek alternatives to back up beepers. The Mine Safety and Health Administration was approached on this issue but their regulations absolutely require the use of audible warning beepers during daylight hours (lights can be used at night). Southdown has adjusted the beepers to be as quiet as possible and still be audible above the background noise of the mining operation.

Closure of Larimer County Quarry

Southdown committed to stop shipments of limestone from its Larimer County Quarry after the startup of Dowe Flats. The final loads of rock were delivered to the plant in January, 1997. Since that time, the Larimer County Quarry has been closed and reclaimed.

Other Permit Requirements

Several State and local permits were required to mine at Dowe Flats. All of these have been applied for and/or obtained. They are:

- Colorado Mined Land Reclamation Board - State Permit for mining and reclamation
- Colorado Department of Transportation - permit for access, crossing, and conveyor tube over State Highway 66
- Colorado Department of Public Health and Environment - air emissions permit, water discharge permit
- Colorado Division of Water Resources - Temporary substitute water supply plan
- Boulder County - Floodplain Development Permit

Status and Plans for the Old Plant Quarry

Although there were no permit requirements regarding the old quarry, it is understood that our neighbors have a natural curiosity to know Southdown's plans for this area. A transition from mining to reclamation is taking place in the existing limestone quarries. A small amount of limestone production will continue at the existing quarries but most of the activity will be reclamation. There have been four (4) pits in operation over the last several years: A, B, C, E Pits.

A-Pit at the southeast corner of the property has been filled as a lake covering about 50 acres, as allowed by a County Special Use Permit and a CMLRB permit. Both were approved in the 1980's.. The surrounding shore areas will be graded and planted over the next 18 months.

B-Pit (at the southwest corner near Hygiene Road) has been a Cement Kiln Dust (CKD) disposal pit for the last 10 years. It will be closed and reclaimed within the next 18 - 24 months.

C-Pit at the center of the old quarry will remain as an active mine pit and will be used for CKD disposal over the next 25 years.

E Pit, south of Hygiene Road, has been completely backfilled with overburden and topsoiled. It will be seeded with winter wheat and dryland pasture grasses this fall. This pit is the last location south of Hygiene Road to be mined and reclaimed by Southdown. This property will be sold to Mr. James Guercio this fall and will become part of his Caribou Ranch.

MANAGEMENT AND MONITORING PLANS

The completed Dowe Flats Project Management and Monitoring Plans were submitted to Boulder County on September 12, 1994. The following discussion reviews each of the domains from this document and incorporates other relevant events and information related to the environmental studies for this project that have taken place since the year one review.

Wildlife

An interim report, summarizing field work completed during the fourth year of the five year winter raptor study, has been submitted to Boulder County. Highlights of the report include:

- Field work conducted this past winter sampled transitional activities in both study areas. At the existing Lyons quarries, mining was being completed and large scale reclamation activities were underway. In Dowe Flats, quarry, conveyor and crusher construction was taking place during the entire study period.
- The prairie dog and lagomorph (cottontails and jack rabbits) population and distribution continues to gradually increase following the plague epizootic of 1994.
- Large raptor use in both study areas continues to track the prairie dog prey base.

Data associated with the effects of mining and other human disturbances on winter raptors continues to be collected. The field work for the entire study will be completed in April of 1998. A comprehensive statistical analysis will follow, and a final report will be submitted later in 1998.

Dowe Flats American Indian Advisory Council

The last meeting was held in November of 1995. No meeting was held in 1996 since no major project construction had taken place at that time. The next meeting is scheduled for October 31 and November 1 of this year. The agenda will include a field tour of the mining site, discussion of project construction, reports on the field monitoring of topsoil in the mining area and archaeological sites outside of the mining area, and management of the Indian Mountain property.

Archaeology

Monitoring of topsoil removal in the mining area for prehistoric resources is a requirement of the project Cultural Resources Management Plan. The project field monitor is Walker Nightwalker. He was selected by the plant manager and approved by the Dowe Flats American Indian Advisory Council in 1995. Michael Burney is the backup monitor. Both field monitors have served in these roles since the beginning of the project. Monitoring logs are filed with the plant manager.

Annual monitoring of archaeological sites outside of the mining area on Southdown's buffer properties was conducted in 1996 and 1997 by Marcus Grant, the project archaeologist. The 1996 monitoring report was submitted to Boulder County. The 1997 report is currently being completed.

Historic

As part of a county-wide recordation of historic structures, the Harroun Farm/Garner Dairy was inventoried by cultural resource specialists. Portions of the site were recommended as eligible for nomination to the National Register of Historic Places (NRHP). The Dowe Flats project is anticipated to have minimal impact on this site since the mine operations center was moved from this site to a location approximately one-half mile to the west.

The Montgomery School, an off-site NRHP listed resource, has been monitored by the project historian. The monitoring for 1997 found no change in the condition of the site. Seismographic monitoring during blasting in the mining area found no impacts (zero readings) from blasting. A 1997 monitoring report has been submitted to Boulder County.

The project historian is also responsible for periodic updates of the project Cultural Resources Management Plan (CRMP). An updated addendum has been submitted to Boulder County. This addendum contains additional details on the archaeology, the Dowe Flats American Indian Advisory Council, and historic domains.

Wetlands

Wetland boundaries were field delineated and marked prior to project construction in Dowe Flats. These boundaries were field checked after mining construction and no impacts to wetlands were documented.

Subsequent to the relining of portions of the St. Vrain Supply Canal in the northwest section of Dowe Flats, additional field analysis was completed on selected wetlands. The Army Corps of Engineers (ACOE) concurs with the project consultant that hydrology has been lost on three wetlands. Other wetlands previously recorded continue to maintain their original status. Copies of the wetlands work and ACOE ruling are on file with the plant manager and Boulder County.

Land Use

Southdown is currently in the process of conveying the "West Dowe Flats Donation Parcel" and the "Northwest Dowe Flats Property" to Boulder County. It is anticipated that this conveyance will be completed by September 30, 1997.

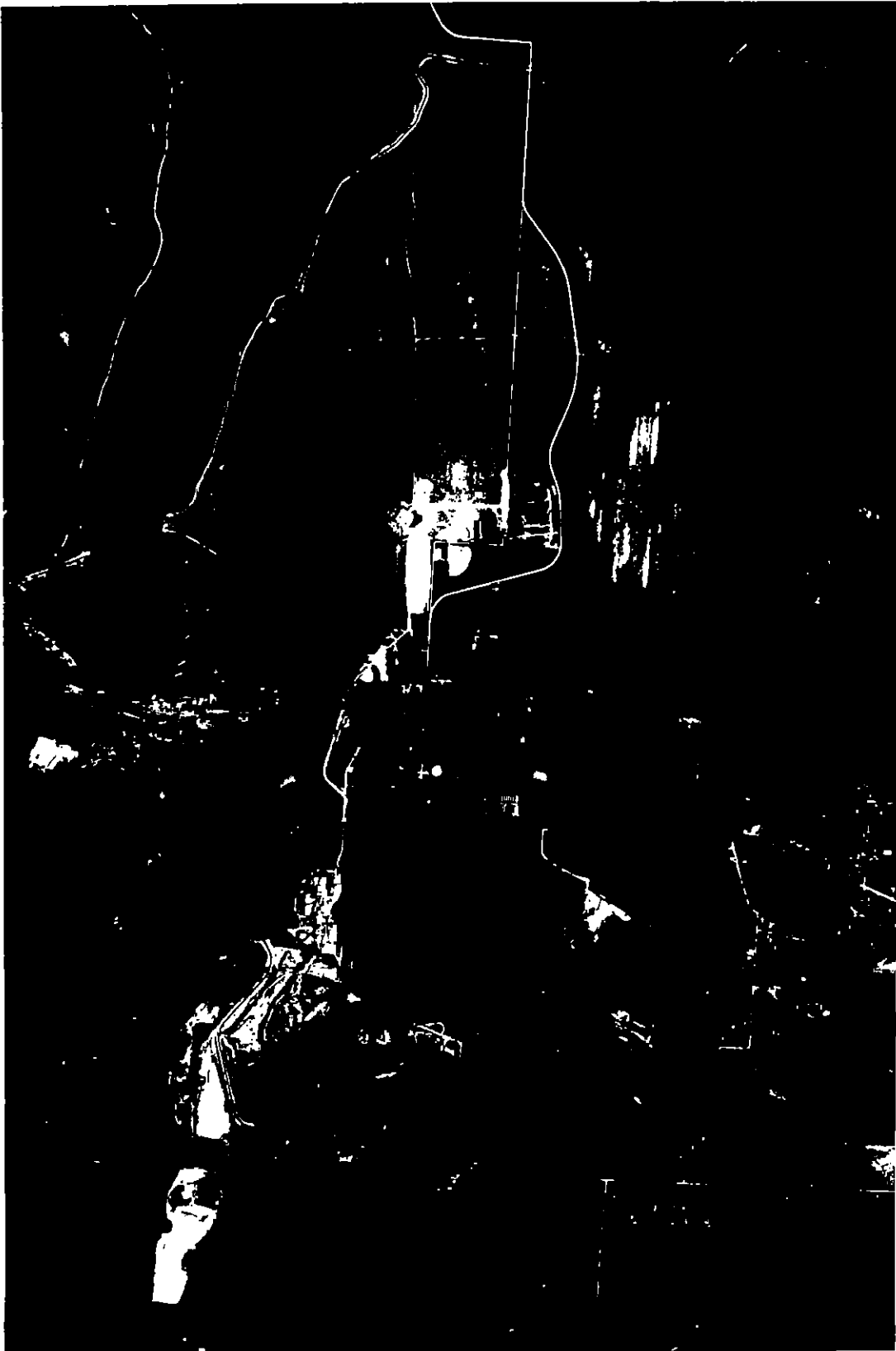
The mine operations center has been moved from the Harroun Farm site (Garner Dairy) to an area adjacent to the southwest corner of the mine quarries. This change was approved by the Board of County Commissioners earlier this year.

Agriculture

No changes in the agricultural operations have been made.



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